Status	Finished
Started	Sunday, 11 October 2026, 8:23 PM
Completed	Sunday, 11 October 2026, 8:34 PM
Duration	10 mins 44 secs
Grade	**12.00** out of 15.00 (**80**%)

Question **1**

Complete

Mark 1.00 out of 1.00

What type of problems can sorting help solve efficiently?

- a. All of the above
- b. Duplicates
- c. Searching
- d. Selection

The correct answer is: All of the above

Question **2**

Complete

Mark 0.00 out of 1.00

Very slow way of sorting is_____

- a. Insertion sort
- b. Bubble sort
- c. Heap sort
- d. Quick sort

The correct answer is:
Insertion sort

Question **3**

Complete

Mark 1.00 out of 1.00

In the context of sorting, what does the divide-and-conquer approach involve?

- a. Sorting data in a single pass
- b. Sorting data sequentially
- c. Rearranging data without sorting
- d. Dividing the input into parts, solving each part, and combining the solutions

The correct answer is: Dividing the input into parts, solving each part, and combining the solutions

Question **4**

Complete

Mark 1.00 out of 1.00

Which built-in Python function is used to sort data?

- a. sort()
- b. arrange()
- c. order()
- d. sorted()

The correct answer is: sorted()

Question **5**

Complete

Mark 1.00 out of 1.00

Why is sorting important for selection operations?

- a. It slows down the process
- b. It makes it easier to select items based on their relationship to the rest of the items
- c. It makes the data unsorted
- d. It complicates the selection of items

The correct answer is: It makes it easier to select items based on their relationship to the rest of the items

Question **6**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm is known for its simplicity and sequential comparison of elements?

a. Quick Sort
b. Merge Sort
c. Heap Sort
d. Bubble Sort

The correct answer is: Bubble Sort

Question **7**

Complete

Mark 0.00 out of 1.00

Two-way merge sort algorithm is used to sort the following elements in ascending order.
200,470,150,80,90,40,400,300,120,70
What is the order of these elements after second pass of the merge sort algorithm?

a. 40,80,90,150,200,300,400,470,70,120
b. 40,70,80,90,120,150,200,300,400,470
c. 200,470,80,150,40,90,300,400,70,120
d. 80,150,200,470,40,90,300,400,70,120

The correct answer is:
80,150,200,470,40,90,300,400,70,120

Question **8**

Complete

Mark 1.00 out of 1.00

_____ explain how an algorithm will perform when the input grows larger.

a. Merging
b. Complexity
c. Searching
d. Sorting

The correct answer is:
Complexity

Question **9**

Complete

Mark 1.00 out of 1.00

Which Python function would you use to sort a list in-place?

- a. sorted()
- b. order()
- c. sort()
- d. arrange()

The correct answer is: sort()

Question **10**

Complete

Mark 0.00 out of 1.00

Which algorithm is efficient for analyzing the frequency distribution of items in a list?

- a. Quick Sort
- b. Bubble Sort
- c. Linear Search
- d. Merge Sort

The correct answer is: Merge Sort

Question **11**

Complete

Mark 1.00 out of 1.00

What is Bubble Sort known for?

- a. Sorting data in a non-sequential manner
- b. Bubbling up the largest element to its correct position with each pass
- c. Using the divide-and-conquer approach
- d. Being the most efficient sorting algorithm

The correct answer is: Bubbling up the largest element to its correct position with each pass

Question **12**

Complete

Mark 1.00 out of 1.00

What is a significant characteristic of Bubble Sort?

a. It uses the heap data structure
b. It recursively sorts subproblems
c. It bubbles up the largest element in each pass
d. It divides the list into sublists

The correct answer is: It bubbles up the largest element in each pass

Question **13**

Complete

Mark 1.00 out of 1.00

What is one advantage of sorting a list before performing a search operation?

a. It makes the search operation slower
b. It has no effect on the search operation
c. It increases the number of comparisons needed
d. It allows for faster searching

The correct answer is: It allows for faster searching

Question **14**

Complete

Mark 1.00 out of 1.00

In Merge Sort, what happens after the two halves of the list are sorted?

a. They are combined to form a single sorted list
b. They are split again into smaller sublists
c. They are compared element by element
d. They are discarded

The correct answer is: They are combined to form a single sorted list

Question **15**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm involves comparing elements and swapping adjacent items that are out of order?

a. Merge Sort
b. Linear Search
c. Bubble Sort
d. Binary Search

The correct answer is: Bubble Sort